

Income Inequality

Contemporary Economic Issues: ECON 204

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Winter 2025

What is Income Inequality?

Income inequality refers to the unequal distribution of income among individuals or households within a country. In Canada, income inequality has been a growing concern, with disparities between the rich and the poor widening over time.

Key Definitions

- **Income Inequality:** The gap between the rich and the poor in terms of income distribution.
- **Gini Coefficient:** A measure of income inequality ranging from 0 (perfect equality) to 1 (perfect inequality).
- **Lorenz Curve:** A graphical representation of income distribution.
- **Quintiles:** Dividing the population into five equal groups based on income.
- **Disposable Income:** Income after taxes and transfers.

Income Inequality in Canada: An Overview

- **Trends:** Income inequality in Canada has increased since the 1980s.
- **Key Statistics:**
 - Top 10% of earners hold about **25%** of total income.
 - Bottom 50% of earners hold about **20%** of total income.
- **Regional Disparities:** Higher inequality in Ontario and Alberta compared to Atlantic Canada.

Gini Coefficient

The Gini coefficient is a numerical measure of income inequality. It is calculated using the Lorenz curve.

$$\text{Gini Coefficient} = \frac{\text{Area between Lorenz curve and equality line}}{\text{Area under equality line}}$$

Income Distribution Table: Example

Consider another income distribution for a hypothetical country:

Income Group	Income Share (%)
Bottom 20%	10
Next 20%	15
Middle 20%	20
Next 20%	25
Top 20%	30

This table shows a more equal distribution of income compared to the previous example.

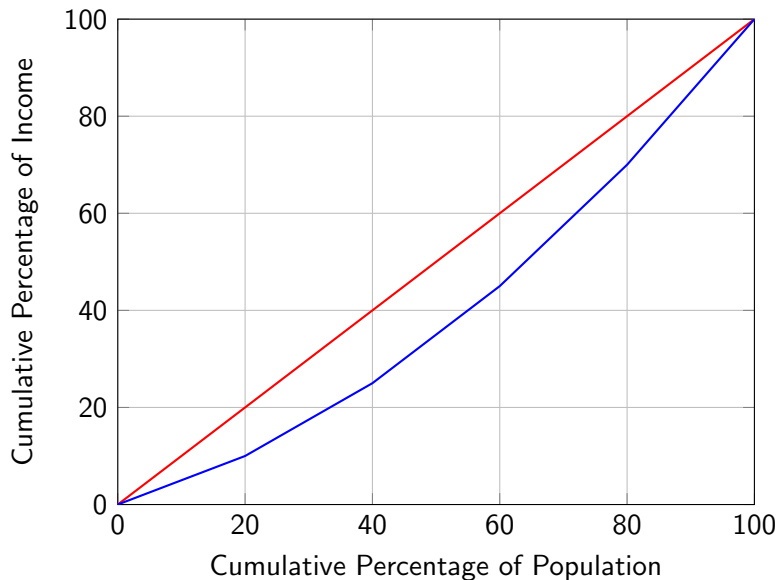
Quintiles and Cumulative Shares: Example

Using the income distribution table, we calculate the cumulative income shares for each quintile:

Quintile	Income Share (%)	Cumulative Income Share (%)
Bottom 20%	10	10
Next 20%	15	25
Middle 20%	20	45
Next 20%	25	70
Top 20%	30	100

The cumulative income share is calculated by adding the income share of each quintile to the previous total.

Lorenz Curve: Example



Trapezoidal Rule

The trapezoidal rule is a method for approximating the area under a curve by dividing it into trapezoids. For the Lorenz curve, we use the trapezoidal rule to calculate the area under the curve.

$$\text{Area of a trapezoid} = \frac{1}{2} \times (\text{Base}_1 + \text{Base}_2) \times \text{Height}$$

In the context of the Lorenz curve: - Base_1 and Base_2 are the cumulative income shares at two consecutive points. - Height is the width of the interval (20% in this case).

Area Calculation: Step 1

1. The area under the line of equality (perfect equality) is:

$$\text{Area}_{\text{equality}} = \frac{1}{2} \times 100 \times 100 = 5000$$

This represents the total area under the 45-degree line.

Area Calculation: Step 2

2. The area under the Lorenz curve is calculated using the trapezoidal rule:

$$\text{Area}_{\text{Lorenz}} = \frac{1}{2} \times (10 + 25) \times 20 + \frac{1}{2} \times (25 + 45) \times 20 + \frac{1}{2} \times (45 + 70) \times 20 + \frac{1}{2} \times (70 + 100) \times 20 = 3900$$

$$\text{Area}_{\text{Lorenz}} = 350 + 700 + 1150 + 1700 = 3900$$

Area Calculation: Step 3

3. The area between the Lorenz curve and the line of equality is:

$$\text{Area}_{\text{between}} = \text{Area}_{\text{equality}} - \text{Area}_{\text{Lorenz}} = 5000 - 3900 = 1100$$

This area represents the degree of income inequality.

Area Calculation: Step 4

4. The Gini coefficient is:

$$\text{Gini Coefficient} = \frac{\text{Area}_{\text{between}}}{\text{Area}_{\text{equality}}} = \frac{1100}{5000} = 0.22$$

A Gini coefficient of 0.22 indicates low income inequality.

Lorenz Curve: Example

Area Calculation: 1. The area under the line of equality (perfect equality) is:

$$\text{Area}_{\text{equality}} = \frac{1}{2} \times 100 \times 100 = 5000$$

2. The area under the Lorenz curve is calculated using the trapezoidal rule:

$$\text{Area}_{\text{Lorenz}} = \frac{1}{2} \times (10+25) \times 20 + \frac{1}{2} \times (25+45) \times 20 + \frac{1}{2} \times (45+70) \times 20 + \frac{1}{2} \times (70+100) \times 20 = 3900$$

3. The area between the Lorenz curve and the line of equality is:

$$\text{Area}_{\text{between}} = \text{Area}_{\text{equality}} - \text{Area}_{\text{Lorenz}} = 5000 - 3900 = 1500$$

4. The Gini coefficient is:

$$\text{Gini Coefficient} = \frac{\text{Area}_{\text{between}}}{\text{Area}_{\text{equality}}} = \frac{1500}{5000} = 0.3$$

Causes of Income Inequality in Canada

- **Globalization:** Outsourcing of manufacturing jobs.
- **Technological Change:** Automation and AI replacing low-skilled jobs.
- **Education Gap:** Unequal access to quality education.
- **Tax Policies:** Reduction in progressive taxation.
- **Wealth Accumulation:** The rich benefit more from capital gains.

Consequences of Income Inequality in Canada

- **Economic:** Reduced economic growth due to lower consumption.
- **Social:** Increased crime rates and social unrest.
- **Health:** Poorer health outcomes for low-income groups.
- **Political:** Concentration of political power among the wealthy.

Policy Responses to Income Inequality in Canada

- **Progressive Taxation:** Higher tax rates for higher income brackets.
- **Minimum Wage Laws:** Ensuring a living wage for low-income workers.
- **Education Reform:** Improving access to quality education.
- **Social Safety Nets:** Expanding unemployment benefits and healthcare.
- **Wealth Taxes:** Taxing inherited wealth and capital gains.

Case Study: Income Inequality in Ontario

- **Key Facts:**

- Ontario has the highest income inequality among Canadian provinces.
- Top 1% of earners hold about **12%** of total income.

- **Causes:**

- Decline in manufacturing jobs.
- High cost of living in cities like Toronto.

- **Policy Responses:**

- Higher minimum wage.
- Expansion of childcare subsidies.

- **Automation:** Widening the gap between high-skilled and low-skilled workers.
- **Universal Basic Income (UBI):** A potential solution to job displacement.
- **Climate Change:** Disproportionately affecting low-income populations.

Conclusion

Income inequality in Canada is a multifaceted issue with significant economic, social, and political implications. Addressing it requires a combination of progressive taxation, social safety nets, and investments in education and training.